Project Guidelines - Work Smarter Not Harder

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Project Guidelines

Great Idea overall! Implementing an actual ML solution and not immediately jumping to a pre-trained GPT model is a great way to work and you will earn a lot from it. because the project is already so detailed and will take a lot from you, I will not add a lot more requirements, but considering how in the proposal there will be no need for parallelization in your project, I will ask you to add a few more features for your project, which will make it complete with all we will learn this semester. Hopefully these features make sense with the vision you had for the project, if they don't, feel free to change to another similar feature that will test the same concepts. But **Let me know via mail/whatsapp if you want to change these guidelines, it is ok to change them but I need to know about it.**

IMPORTANT NOTE: these are the guidelines needed for a project to be at a score of 100, you do not have to complete them all and we do not expect you to complete them all. Remember that each bug we find is -5 points loss and each week of delay in the submission is -5 points loss. It is completely reasonable to only complete some of the features in the guidelines. You will get partial credit for a partial completion of each feature.

The guidelines you have set yourself are as follows:

- **75 points - Project Proposal:** Implement a working, localhost web application that implements your proposed idea, it should include all that was mentioned in the proposal, and you will be checked accordingly.
- **Job Queue, +5 points:** Implement a job queue for the AI container, such that the AI container will be able to handle MANY requests at once from many users, and will be able to process them in parallel.
- **Online Forum, 10 points:** Implement an Online forum page, kind of similiar to a social network/blog (take inspiration from Linkedin). People on this page could through their client to share their milestones and their progress etc. you will have the following features for the online forum (feel free to add/change slightly):
    1. Each client should be able to post on the forum, and each post should be visible to all the other clients. Each post should have a title, a body, and the option to add images and videos to the post as well.
    2. Each client should be able to comment on each post, and each comment should be visible to all the other clients. Each comment should have a body, and the option to add images and videos to the comment as well.
    3. Each client should be able to like/dislike each post and each comment, and see how many likes each post and comment has, he should also be able to look at his posts on his personal area, and see how many likes and dislikes he has overall.

4. There should be a direct-messaging feature between the clients, where each client can send a message to any other client, and the message should be visible only to the sender and the receiver. Each message should have a body, and the option to add images and videos to the message as well.

5. The clients will get a notification on the webapp when they receive a new direct message, they will also be able to see who liked/disliked their posts and comments.

6. **IMPORTANT:** You should think about how to protect your system againts spamming (for example, a user sending 1000 messages in a short time to overload the server), and against sending huge video files to overload the database.

7. **Cold seeding:** to get the ball rolling, the website should be created with already a few "fake" clients, with a few posts and comments on the forum.

This can be implemented in any way you want, and you can add and change the proposed features as you wish (let me know if you are doing a major change) but it should be possible to send messages on this forum and receive them in live feedback (no page refresh should be necessary), the chats should be saved and retrievable for the users, and each user should get to access their chat history and get a live notification in the webapp for a new message. The users have a lot of freedom here, so make sure your server cannot be easily crashed by malicious users, and that you have a good security system in place for this online forum.
10 points for implementing the feature, partial scoring will be given for partial implementation.

- **10 points - Website Deployment & CI/CD:** Deploy your website to an azure cloud server (we will supply you with a server and a domain if you choose to do implement this feature, but you have to implement the deployment yourself). The website should be accessible to the public and be able to scale easily (use parallelization in your code). Implement a CI/CD pipeline for your project's private GitHub repository - the pipeline should include a test set that will run on each commit and check that all the tests pass - if they do, the pipeline should automatically deploy the new version of the website to the server. (We will learn how to do this in lesson 10, but feel free to read up and do it yourself before hand, or ask me for help if you want to implement it before lesson 10 and need my assistance). **Only implementing the CI/CD pipeline without website deployment (only checking that the tests pass on each commit) will award you partial credit of 5 points.**